

HOW IMPORTANT ARE SHARED PERCEPTIONS OF PROCEDURAL JUSTICE IN COOPERATIVE ALLIANCES?

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Building on the “loose coupling” logic, this study examines the importance of boundary spanners’ shared perceptions of procedural justice in cross-cultural cooperative alliances. I argue that alliance profitability is higher when both parties perceive high rather than low procedural justice. Profitability is also higher when both parties’ perceptions are high than when one party perceives high procedural justice but the other perceives low procedural justice. Shared justice perceptions become even more important for alliance profitability when the cultural distance between partners is high or when the industry of operation is uncertain. Analysis of 124 cross-cultural alliances in China supports these propositions.

Procedural justice and its influence on the attitudes, behavior, and performance of individuals, especially employees, have been richly studied in the field of organizational justice. Studies have demonstrated that justice has a positive effect on employees’ trust and social harmony with decision makers (e.g., Korsgaard, Schweiger & Sapienza, 1995; Naumann & Bennett, 2000), commitment to decisions or policy execution (e.g., Masterson, Lewis, Goldman, & Taylor, 2000; McFarlin & Sweeney, 1992), citizenship behaviors (e.g., Ball, Treviño, & Sims, 1994; Konovsky & Organ, 1996; Niehoff & Moorman, 1993), and contribution to organizational change and strategic planning (e.g., Folger & Konovsky, 1989; Mishra & Spreitzer, 1998). In the examination of these issues, perceived procedural justice has been regularly conceptualized and measured from a one-sided perspective or based on unilateral judgments by a single side or party, leaving the importance of two-sided or two-party agreement in justice perceptions largely unexplored.

Procedural justice is important to cooperative relationships between exchange parties, but in my view only when *all* exchange parties perceive its existence can justice really serve as a foundation for such relationships. If justice perceptions are not common to all parties, conflicts and related opportunistic behaviors may arise, since one party is likely to feel unfairly treated with respect to exchange procedures (Klein, Conn, Smith, & Sorra, 2001). This perception may in turn impair cooperation, undercut commitment, or hamper reciprocal

exchange. To address the importance of shared perceptions of procedural justice, this study examined cross-cultural, two-party cooperative alliances structured as separate entities in an emerging market as the conceptual and empirical setting. The parties to each alliance were from different countries, and each party deployed critical resources to its joint operation in search of certain shared goals or collective payoffs. More specifically, I attempted to determine whether the level of perceived procedural justice commonly agreed upon by both parties (*shared procedural justice* hereafter), through their boundary spanners (each party’s senior representatives in an alliance), was positively related to an alliance’s performance (profitability). Further, I examined whether shared procedural justice had a stronger positive relationship with alliance profitability than the level of procedural justice unilaterally perceived by each individual party (*unilateral procedural justice* hereafter). Here, the level of shared procedural justice is an interaction term defined as the mean of two individual parties’ perceived procedural justice scores, multiplied by the interparty agreement estimate (r_{wg}). Collective perceptions of procedural justice may involve agreement or disagreement and high or low perceptions of procedural justice.

Cross-cultural alliances are characterized by cultural differences between the investing parties, and by the uncertainties they face (Park & Ungson, 1997; Parkhe, 1993). Given that, I also intended to scrutinize whether the positive effect of shared procedural justice was stronger when interpartner cultural distance was higher and when alliances operated in more uncertain industries. To address the above issues, I built on the loose coupling logic (Orton & Weick, 1988, 1990), supplementing this

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logic with convergent insights from justice theory and alliance theory. According to this logic, an alliance is a “loosely coupled system” in which parties maintain their respective identities while interdependently sharing resources, an arrangement that often causes factual disagreement in justice perceptions. Shared justice perceptions help hold together such a system, and serving this function is likely to be especially important when large cultural differences between parties and high uncertainty exaggerate looseness.

SHARED JUSTICE AND COOPERATION

Theoretical Background

What justice entails varies according to its organizational context and level (Greenberg, 1987; McFarlin & Sweeney, 1992; Schminke, Cropanzano, & Rupp, 2002). In the context of cooperative alliances, I define procedural justice as the fairness of an alliance’s strategic decision-making process and the procedures that influence each party’s gains and interests, as perceived by the boundary spanners who represent each party. In a general setting, procedural fairness means that the procedures and criteria used in making and executing decisions are unbiased, representative, transparent, correctable, and ethical (Leventhal, 1980). In an alliance setting, procedural fairness entails not only all the elements relevant in a general setting, but also congruency with the terms and specifications of the alliance’s contract or agreement.¹ I view an alliance contract as one important standard by which to judge fairness during interpartner exchange. Fairness is required in procedures for (1) building and structuring an alliance (i.e., board formation and decision making, contract codification, and alliance establishment), (2) organizing and managing an alliance (i.e., strategic planning and routine management), (3) governing resource sharing (i.e., knowledge transfer, innovation and resource contribution), and (4) executing alliance plans and decisions (i.e., clarity of execution procedures, contract-execution monitoring and decision-execution monitoring). Procedural justice is judged by boundary spanners whose perceptions of fairness affect the parent firms’ actions. Boundary spanners are

typically the two chief managers (top executives) who represent the parties to the alliance (Luo, 2001; Shenkar & Zeira, 1992). The ties between these executives play a major role in the maintenance of interorganizational relationships (Levinthal & Fichman, 1988; Seabright, Levinthal, & Fichman, 1992) and are essential to joint ventures’ overall performance (Johnson, 1997; Luo, 2001). Because their perceptions of fairness are affected by multiple factors, including each parent firm’s strategic intent and bargaining power as well as each boundary spanner’s experience and knowledge, boundary spanners often have differing perceptions of procedural justice.

The importance of shared procedural justice perceptions is underpinned by the loose coupling logic (for a review, see Orton and Weick [1990]). A cooperative alliance is a loosely coupled system in which parties interdependently share existing resources or jointly develop new resources (*coupling*) while maintaining their respective identities, resource control, and parent-alliance separation (*looseness*). Thus, loose coupling is a situation in which elements are interconnected and mutually dependent but retain separateness and identity. Loose coupling suggests that any location in a coalition contains interdependent elements that vary in the number and strength of their interdependencies. The result is a system that is simultaneously open and closed, indeterminate and rational, spontaneous and deliberate, and “loose” and “coupled” (Luke, Begun, & Pointer, 1989; Orton & Weick, 1988). Looseness, arising from each party’s freedom to adjust its commitment and maintain its identity, is likely to be the main structural reason for interparty disagreement on justice. While it engenders flexibility, looseness allows each party to perceive, assess, and appraise procedural justice (via boundary spanners) from its own perspective. In such a loose system, each party bases its resource contributions and organizational commitments to alliance activities on both contractual stipulations and its own assessment of justice. Given that different parties and their boundary spanners have asymmetrical sources of information and different abilities to search, collect, scan, and analyze information (Jones, Hesterly, & Borgatti, 1997; Tushman & Nadler, 1978), looseness is in a way responsible for the justice disagreements that arise from these asymmetries. In a cross-cultural setting, different parties (and their boundary spanners) have varying thresholds and anticipations regarding justice. They may interpret it idiosyncratically, owing to differing cultural backgrounds or cooperation experience (Farh, Earley, & Lin, 1997; Pillai, Scandura,

¹ In this study I at times use justice and fairness interchangeably for the same concept. Since there is a growing consensus that procedural justice and interaction justice are separate concepts (e.g., Bies & Moag, 1986), I did not include interaction justice elements (social and interactional aspects of decision implementation) in the definition and measurement of procedural justice.

& Williams, 1999), which may foster even more disagreement on procedural justice.

My suggestion is that, together with the mean level of perceived procedural justice, sharing procedural justice perceptions is important to the stability and sustainability of coupling. According to Orton and Weick (1990), the "sharedness" of certain fundamental beliefs is the critical force that holds together a loosely coupled system. They explicitly stated that "if organizations are determinate means-ends structures for attaining preferred outcomes, and if loose coupling is produced by uncertainties about these means-ends structures, then agreement about values and preferences is the only source of order that is left" (Orton & Weick, 1990: 212). Although Orton and Weick did not detail what shared beliefs comprise, I propose that shared procedural justice perceptions represent one such fundamental belief, and as such, act as a cohesive force that fosters interparty cooperation. Specifically, when shared justice perceptions between boundary spanners are low, the role of justice in governing the order of interfirm exchange or reducing the level of interfirm conflict may decline. When a disagreement occurs, the party that rates justice lower will perceive less fairness in cooperation procedures, which in turn will plant the seeds for disputes in various areas such as managerial control, strategic planning, and joint venture development. Johnson (1997) demonstrated that a party that perceives low procedural justice tends to deliver low commitment to joint venture operations. In addition, asymmetry in justice judgments implies attitudinal bias and heterogeneous impetuses for action, which may disrupt the process of reciprocation, learning, and resource sharing. Shaw (1981) suggested that such bias hinders group cohesion. Hill (1990) argued that heterogeneous impetuses for action will lead to opportunism in joint ventures unless the pursuit of self-interest is suppressed.

In a coupled situation underpinned by shared procedural justice, confidence and reciprocation from each party are likely to increase. These factors, together with ceremonial management and shared views on tasks, actions, and policies, are the main forces that maintain coupling and may generate more collective gains (Meyer & Rowan, 1977: 358). The relational model in justice theory suggests that justice facilitates neutrality and standing, which in turn improve reciprocation and communication (Allen & Meyer, 1990; Brockner & Siegel, 1995). I expect that, if both parties to an alliance hold common and high justice perceptions, reciprocation and communication may be further improved because this common perception encour-

ages openness in interactions and curbs fears of exploitation.

Main Effects of Shared Procedural Justice

The argument here is that the level of shared procedural justice perceived by two parties to an alliance is likely to maintain a greater positive relationship with alliance profitability than the level of procedural justice perceived by each individual party. This proposition entails two conjectures: (1) there is a positive link between the level of shared procedural justice and profitability (that is, profitability is higher when both parties perceive high procedural justice than when both parties perceive low procedural justice), and (2) there is a stronger positive link between the level of shared procedural justice and profitability than between the level of unilaterally perceived procedural justice and profitability (that is, profitability is higher when both parties perceive high procedural justice than when one party perceives high procedural justice but the other party perceives low procedural justice, even if the sum of both parties' judgments of procedural justice is the same in the two cases).

The first conjecture is rooted in the logic that a high level of shared procedural justice improves process efficiency and reduces administrative costs, thus increasing financial returns. Procedural justice creates standards and norms of expected behavior that not only obviate the need for, but are superior to, pure authority relations in discouraging malfeasance in operations and management (Tyler, 1989). Such standards and their generalized morality can strengthen process control in production, operations, marketing, and organization (Wathne & Heide, 2000). A work group's cohesion resulting from higher shared procedural justice further improves procedural justice climate and process efficiency (Naumann & Bennett). Heightened commitments from each party resulting from improved procedural justice also support business strategy implementation (Kim & Mauborgne, 1993) and streamline joint operations (Johnson, 1997). With respect to cost savings, increased procedural justice as commonly perceived by boundary spanners promotes coordination and routinization for alliance activities, which collectively reduce operating and managerial costs. In addition, higher shared procedural justice reduces interpartner differences in corporate culture, strategic orientation, and managerial style, further promoting cost reduction and managerial efficiency. Finally, cooperative alliances are characterized by internal instability (Kogut, 1989), but this instability is at least somewhat mitigated by higher shared procedural justice

because boundary spanners' shared fairness perceptions strengthen the embeddedness of formalized or routinized procedures and policies (Johnson, 1997). Moreover, increased stability further saves coordination and control costs for joint activities (Killing, 1983; Mjoen & Tallman, 1997). In light of the above discussion, I propose:

Hypothesis 1a. There is a positive relationship between shared procedural justice and alliance profitability.

My second conjecture—that there is a stronger positive link between the level of shared procedural justice and profitability than between the level of unilaterally perceived procedural justice and profitability—is based on the logic that shared procedural justice adds more value to financial returns than does unilateral procedural justice, even when the sums of two parties' unilateral and shared judgments are the same. It is true that high unilateral procedural justice implies that one party to an alliance rates fairness as high, and at least this party may become highly committed to joint operations. In this sense, high unilateral procedural justice may partially contribute to alliance performance. Beamish and Banks (1987) found that unilateral commitments, especially by local partners, can improve cross-cultural alliances' local acceptance and networking.

One party's favorable perception of procedural justice and its commitment to an alliance are not sufficient to establish a sustained healthy relationship and to achieve superior joint returns. A one-sided perception is not adequate to counteract conflicts or managerial incongruence that emerge as the alliance develops. In contrast to unilateral procedural justice, shared procedural justice may *bolster* alliance profitability via improved dyadic relationships, and thus strengthen a positive link between shared procedural justice and profitability. First, commonly agreed-on procedural justice may create more relational value for joint operations than unilaterally perceived procedural justice. *Relational value* is a collective gain arising from the complementary aligning of resources and commitments in a cooperative exchange (Das & Teng, 1998). When both parties perceive high procedural justice, their individual actions and commitments are more likely to be harmonious and consistent with collective goals. This harmony favors synergy creation and revenue growth (Dyer, 1997) and saves costs in interpartner coordination (Gulati, 1995). Second, shared procedural justice may more effectively curtail relational risks than unilaterally perceived procedural justice. *Relational risk* is the probability that partners will not

cooperate fully and will behave in some opportunistic manner (Das & Teng, 1998). Joint profitability always suffers from opportunism (Hill, 1990). A high level of shared procedural justice reduces bargaining expenses because formalized procedures are perceived as unbiased toward both parties. In this situation, each party is motivated to contribute more resources to the joint operation and is more willing to share knowledge with the partner firm. Finally, a high level of shared procedural justice helps increase interpartner conformity in strategic responses to market changes, even if the consequences of such changes are unfavorable to one party. Lind and Tyler (1988) suggested that the convergence of group values and the agreement in fairness judgments countervail the "frustration effect." Shared procedural justice hence reduces interparty conflicts and improves communication and openness. Beamish and Banks (1987) documented that reduced conflicts improve alliances' financial returns. Accordingly, I posit:

Hypothesis 1b. There is a stronger positive relationship between shared procedural justice and alliance profitability than between unilaterally perceived procedural justice and alliance profitability.

Moderating Effects of Culture and Uncertainty

I also propose that the above effects are even more salient when cultural distance between partner firms is large or when alliances operate in uncertain industries. Cultural distance and structural uncertainty both tend to increase the looseness in a loosely coupled system; the parties are more "separated" when cultural differences or expected uncertainty are great (Watson, Kumar, & Michaelsen, 1993). Heightened separateness in turn compels a stronger need for parties to share common views as a means of holding the system together (Orton & Weick, 1990). As previously discussed, cultural distance and structural uncertainty attenuate coupling; in compensation, shared procedural fairness perceptions become more imperative in sustaining interparty exchanges. Given improvement in shared procedural justice, the members of an alliance will increase their commitment to it even when the outcome is unfavorable under uncertainty (Greenberg, 1987; Lind & Tyler, 1988).

Specifically, I posit that shared procedural justice will be even more important to interparty exchange if the cultural distance between two parties is large because common views on existing fairness can bind the two culturally distant firms together during the course of cooperation. For example, for-

bearance and reciprocity are two critical conditions necessary for alliance evolution and alliance profitability (Dyer, 1997; Parkhe, 1993), but they are disrupted when the cultural backgrounds of the parties differ substantially (Shenkar & Zeira, 1992). However, when the parties commonly perceive high procedural justice, the disruption will be lessened because the mutually agreed upon impartiality pertaining to exchange procedures counters the negative effect of a cultural clash on confidence in the cooperation (Das & Teng, 1998). Research on conflict has shown that shared justice perceptions are essential to the reduction of conflicts arising from cultural differences between parties to an alliance. They must also share justice perceptions if they are to mutually embrace ideas, visions, and philosophies in such a way that a cooperative culture is established and embedded within the alliance (Jehn & Weldon, 1997; Lin & Germain, 1998; Mohr & Spekman, 1994). The greater the cultural distance, the more important are shared justice perceptions to mutual embracing and conflict reduction, and in turn to cost saving and synergy creation. I thus expect:

Hypothesis 2. When cultural distance between partner firms is high, shared procedural justice becomes even more important for alliance profitability.

Uncertainty is also salient to the need for shared procedural justice. Uncertainty increases the separateness in a loosely coupled system (Lutz, 1982). To mitigate this dividing force, shared procedural justice perceptions become essential to sustaining cooperation. When an alliance operates in a highly uncertain industry, the market uncertainty it faces is typically beyond its control (Boyd & Fulk, 1996). In such circumstances, boundary spanners can maintain alliance stability and even improve alliance performance by increasing internal stability and resource sharing, thus reducing their firm's external dependence and alleviating market uncertainty hazards (Kogut, 1989; Pfeffer & Salancik, 1978). As an internal stabilizing force, shared values and beliefs encourage commitment from each party and facilitate reciprocal exchange, despite market uncertainty (Gulati, Khanna, & Nohria, 1994) and, as I suggested earlier, shared procedural justice perceptions represent one such belief. According to the justice literature, people facing uncertainty tend to emphasize procedural fairness in determining their commitment (Brockner & Wiesenfeld, 1996; Kwong & Leung, 2002). Because external uncertainty often accentuates internal conflict between cross-cultural partners (Jehn & Weldon, 1997; Kogut, 1989) and increases the role con-

flict and role ambiguity of boundary spanners (Luo, 2001; Shenkar & Zeira, 1992), shared procedural justice will play a larger role in encouraging cooperative commitment (or discouraging the wait-and-see attitude) and reducing interparty conflicts (or increasing interparty attachment) when high uncertainty is present. All told, heightened commitment, reduced conflict, and increased attachment can enhance an alliance's financial returns (Luo, 2001; Madhok, 1995; Parkhe, 1993). I thus predict:

Hypothesis 3. When structural uncertainty is high, shared procedural justice becomes even more important for alliance profitability.

METHODS

Data and Sample

This study used international cooperative alliances in China to test the above hypotheses. As the world's largest host of foreign direct investment (FDI), China is the primary destination for these alliances. Since 2000, the nation has been attracting an average \$130 million FDI per day, about 65 percent of which stems from cooperative alliances (*Far Eastern Economic Review*, 2002). Justice is considered to be a critical factor affecting Chinese peoples' job satisfaction, productivity, and commitment to organizations (Farh et al., 1997). Prior research has also demonstrated that local managers often judge justice differently than their foreign counterparts, giving rise to disagreements regarding perceived justice (Chen, Choi, & Chi, 2002).

This study uses data collected from three major sources: (1) An initial survey was conducted May–August 2000, targeted to each sample alliance's general manager; each general manager responded to questions regarding such issues as alliance background, the party she or he represented (foreign or Chinese), her or his own justice perceptions, and the name of the leading manager (typically, the senior deputy general manager) of the other party in the alliance. (2) A second survey, conducted right after the first, focused on the second party's justice perceptions; it was sent to the chief manager whose name had been offered by his or her counterpart in the first survey. These two surveys enabled examination of the level of agreement in the procedural justice perceptions of the boundary spanners in each sample alliance. (3) As a third source of data, I used archival information to measure alliance profitability, structural uncertainty, cultural distance, and other control variables. Developed in both English and Chinese, the survey questionnaire was initially cross-checked by four management professors in China and further pre-

tested for instrument validity by 18 alliance managers. Drawing on feedback from the scholars and managers, I reworded questions or terms that I considered ambiguous.

Sample alliances were identified from the *Directory of Foreign-Invested Enterprises* compiled by China's Ministry of Foreign Trade and Economic Cooperation (MOFTEC). I limited the sample to cooperative alliances with two partners only (one foreign and one Chinese), since multiparty alliances (three or more partners) are not very common and can complicate cooperation behavior analysis. The sample was also limited to alliances in manufacturing sectors in Jiangsu Province that had been established for at least three years (most survey questions referred to the past three years). Although using a single province presents some limitations, Jiangsu is appropriate for analyzing foreign alliances for several reasons. First, Jiangsu is second in China in luring foreign capital and third in generating GNP. Second, it has very well developed as well as much less developed regions, mirroring broader disparities across regions of the nation. Third, it treats alliances the same way other provinces do. Fourth, Jiangsu is very representative of the nation's cultural norms and standards. Last, I was able to collect information about alliance profitability from the provincial authority's archival database to help obviate the bias of common method variance. I found no evidence of representative bias in my sample, as compared to the country's total population.

Using the above criteria, I selected 440 alliances in Jiangsu Province as my sample for the first survey. I collected 176 useful questionnaires out of 440 sent, a 40 percent response rate. Of these 176 first-survey sample alliances, 124 completed my second questionnaire, yielding a 70.45 percent response rate with respect to the second survey's distribution (and a 28.18 percent overall response rate). Of the 124 final matched sample alliances, 88 were equity joint ventures and 36 were nonequity joint ventures. Partners in equity joint ventures share ownership (equity) of joint capital investment, whereas nonequity joint ventures instead share noncapital resources such as technology, machinery, and management expertise. The major sources of foreign capital funding these alliances include European Union countries, the United States, Japan, Canada, Australia, Singapore, and Taiwan, among others. These sample alliances participate in various manufacturing industries such as electronics and electric equipment (29), textiles, sewing, and shoes (22), toys and light industry products (15), telecommunications (13), chemical

and plastic products (12), machine building (8), pharmaceuticals (6), and others (19).

Nonresponse bias in the final sample was checked with information from the *Directory of Foreign-Invested Enterprises*. I compared the responding and nonresponding firms along major firm attributes such as total investment, foreign equity, duration, registered capital, and age using *t*-tests. All *t*-statistics were nonsignificant. To verify the representativeness of my final sample, I compared the sample's investment size, registered capital, age, and profitability to the same characteristics of the alliance population nationwide, using information from the *China Statistical Yearbook*. No *t*-statistics were significant, suggesting no significant difference from the population. I also conducted semistructured interviews with 15 first-survey respondents and 9 second-survey respondents to check the accuracy of their responses. The results (Guttman split-half *Rs*, 0.88–0.94 for first-survey respondents; 0.89–0.92 for second-survey respondents) displayed high consistency in their interview and survey answers.

Measurement

My early definition of procedural justice in cooperative alliances, plus several studies on justice (Kim & Mauborgne, 1993; Lind, Kulik, Ambrose, & de Vera Park, 1993; McFarlin & Sweeney, 1992; Tyler, 1989; Welbourne, Balkin, & Gomez-Mejia, 1995) and alliances (e.g., Johnson, 1997), were the bases for development of a multi-item construct to measure procedural justice (1, "not true at all," to 7, "very true"). Procedural justice was measured as the average score of eight items that addressed the fairness of the following over the three years prior to survey administration: (1) the procedures used in board meetings and decision-making processes in these meetings, (2) the procedures the parties used in negotiating, stipulating, and codifying their alliance contract, (3) the procedures the parties used in formulating and structuring their alliance, (4) the procedures used in planning, organizing, and managing alliance activities (e.g., strategic planning, autonomy allocation, and routine management), (5) the procedures used to govern knowledge or resource sharing between the two parties (e.g., knowledge transfer, innovation development, and resource contribution), (6) the procedures for executing strategic decisions, including the clarity of their definition and consistency of their performance, (7) both parties' administration and monitoring of contract execution, and (8) both parties' administration and monitoring of the implementation of strategic decisions. In the questionnaire, I

defined “fair” as referring to whether the above procedures and criteria were unbiased, representative, and transparent, as well as to whether they conformed to the specifications described in the alliance agreement. I also listed some examples to further illustrate how respondents were to understand “unbiased,” “representative,” and “transparent.” For example, I listed the following to explain “transparent”: (1) there are well-defined channels for open discussion between the parties during formulating and executing related procedures; (2) related procedures or joint decisions are always made after consideration of both sides’ views; and (3) it is clear what is expected from both sides regarding preparing, altering, and adopting related procedures. For “unbiased,” I specified that (1) there is no discrimination against either party in preparing and executing related procedures; (2) each party’s participative power in preparing and executing related procedures complies with the alliance contract and is commensurate with its resource commitment to joint operations; and (3) all related procedures are designed for the sake of joint returns and alliance growth, not for an individual party’s sole benefit.

The internal consistency and item appropriateness of the measure of the procedural justice perceptions of the foreign party to each alliance (foreign-party-perceived procedural justice) were respectively validated by Cronbach’s alpha (.79) and communality loadings (.76–.92). For Chinese-party-perceived procedural justice, internal consistency and item appropriateness were also confirmed by Cronbach’s alpha (.74) and communality loadings (.79–.90). Shared procedural justice was measured as the average of the two individual parties’ perceived procedural justice, multiplied by the interparty agreement index ($[procedural\ justice_1 + procedural\ justice_2]/2 \times r_{wg}$), where r_{wg} is James, Demaree, and Wolf’s within-group interrater reliability (1984: 88).

I measured an alliance’s *profitability* by averaging return on investment (ROI; net profit/total investment) over the period 2000–02. The ROI data came from the Jiangsu Provincial Commission of Foreign Trade and Economic Relations. *Structural uncertainty* was defined as the geometric average of standard deviations in the output, sales, and profits of the industry in which an alliance participated ($[\text{s.d. output} \times \text{s.d. sales} \times \text{s.d. net profit}]^{1/3}$); all information was obtained from the *China Statistical Yearbook* for 1998 to 2000. *Cultural distance* was measured via Kogut and Singh’s composite index (1988) and Hofstede’s latest data set (2001: 500–502), which contains five cultural dimensions (power distance, uncertainty avoidance, individu-

ality, masculinity, and long-term orientation). Six control variables were included in the tests: (1) *alliance type* (1 if equity joint venture, 0 otherwise), (2) *size* (investment scale in millions of dollars), (3) *age* (number of years present in China), (4) *location* (1 if a coastal city, 0 otherwise), (5) *market orientation* (1 if export-oriented, 0 otherwise), and (6) *past performance* (average ROI between 1998 and 1999).² Control variables were measured on the basis of information obtained from the *Directory of Foreign-Invested Enterprises*, except for the last control variable, for which I obtained information from the aforementioned commission in Jiangsu Province. Table 1 displays the descriptive statistics and Pearson correlations for every variable in this study.

RESULTS

To examine the relationship between shared procedural justice and alliance profitability, I conducted a hierarchical regression analysis; Table 2 presents these results. After the mean-centering technique had been used, the variance inflation factor estimates (1.4–3.6) for all variables in the full model (model 4) discounted any multicollinearity among them. A modified Kolmogorov-Smirnov test also validated the normality of these variables and the dependent variable ($p > .20$ for each). As shown in the full model, which controls for all related variables and interaction terms, shared procedural justice is significantly associated with alliance profitability ($\beta = .22, p < .01$). This evidence lends support to Hypothesis 1a, suggesting that greater financial returns arise when both parties to an alliance perceive high procedural justice than when both parties perceive low procedural justice. The model 4 coefficients also exhibit some initial evidence that shared procedural justice exerts a stronger influence on alliance profitability than does a foreign partner’s unilateral procedural justice perceptions ($\beta = .05, p > .50$) or a Chinese partner’s unilateral procedural justice perceptions ($\beta = .14, p < .10$).

To perform a finer-grained analysis of the importance of shared procedural justice to alliance profitability compared to that of unilaterally perceived procedural justice, I followed Budescu’s approach (1993), which investigates the relative importance of predictors in hierarchical multiple regression.

² Past performance was included as a control variable since it could affect future performance (Lind & Tyler, 1988) as well as procedural justice perceptions (Brockner & Wiesenfeld, 1996).

TABLE 1
Descriptive Statistics and Pearson Correlations^a

Variables	Mean	s.d.	1	2	3	4	5	6	7	8	9	10	11
1. Shared procedural justice	4.72	1.56											
2. Unilateral procedural justice, foreign	5.63	1.21	.53***										
3. Unilateral procedural justice, Chinese	5.35	1.19	.60***	.43***									
4. Return on investment	15.60	3.94	.28***	.19*	.24**								
5. Structural uncertainty	16.70	5.77	-.05	-.03	-.06	-.12							
6. Cultural distance	5.83	1.47	-.07	-.04	-.02	.04	-.03						
7. Alliance type	0.71	0.45	.10	.08	.03	.07	.04	-.09					
8. Alliance location	0.65	0.47	.11	.06	.10	.18*	-.06	.05	.04				
9. Alliance orientation	0.30	0.46	.17*	.15	.13	.03	-.22**	.07	-.03	.12			
10. Alliance size	7.36	3.92	.04	.05	-.02	-.09	.10	.07	.19*	.23**	.08		
11. Alliance age	9.04	1.97	.14	.24**	.11	.25**	.05	-.04	.09	.12	.03	.19*	
12. Past performance	12.50	2.77	.22**	.14	.18*	.17*	.07	-.03	.11	.20**	-.06	.08	.21

^a $n = 124$.

* $p < .05$

** $p < .01$

*** $p < .001$

TABLE 2
Results of Moderated Hierarchical Regression Analysis^a

Variables	Return on Investment			
	Equation 1	Equation 2	Equation 3	Equation 4
Shared procedural justice		.26***	.25**	.22**
Unilateral procedural justice, foreign		.09	.08	.05
Unilateral procedural justice, Chinese		.17*	.18*	.14 [†]
Structural uncertainty			.06	.04
Cultural distance			.08	.06
Shared procedural justice $\times$ structural uncertainty				.16 [†]
Shared procedural justice $\times$ cultural distance				.18*
Control variables				
Alliance type	.08	.07	.07	.06
Alliance location	-.06	-.06	-.05	-.05
Alliance orientation	.07	.06	.06	.05
Alliance size	-.10	-.08	-.09	-.08
Alliance age	.29***	.27***	.26***	.25***
Past performance	.18*	.18*	.17*	.17*
Model F	6.75*	16.49***	18.64***	22.81
R^2	.15	.30	.31	.40
ΔR^2		.15	.01	.09
Hierarchical F		7.84**	0.73	8.19**

^a The entries in the table are standardized coefficients (β s). The mean-centering technique (Aiken & West, 1991) was used for shared and unilateral procedural justice and for the two interaction terms.

[†] $p < .10$

* $p < .05$

** $p < .01$

*** $p < .001$

Table 3 presents these results. Comparing equation 3 and equation 2 in Table 3 suggests that adding shared procedural justice perceptions ($\beta = .30$, $p < .001$) to foreign-party-perceived justice ($\beta = .14$,

$p < .10$) significantly increases the explanatory power in predicting the variance of profitability ($\Delta R^2 = .05$, hierarchical $F = 8.58$, $p < .001$). Equation 5, which compares shared procedural justice

TABLE 3
Relative Importance of Shared and Unilateral Procedural Justice: Hierarchical Regression Comparison^a

Variables ^b	β	Equation <i>F</i>	R^2	ΔR^2	Hierarchical <i>F</i>	<i>df</i>
Equation 1 Control variables		7.13*	.22			8, 115
Equation 2 (vs.1) Unilateral procedural justice, foreign Control variables	0.17*	7.76**	.24	.02	3.16 [†]	9, 114
Equation 3 (vs.2) Shared procedural justice Unilateral procedural justice, foreign Control variables	0.30*** 0.14 [†]	15.11***	.30	.05	8.58***	10, 113
Equation 4 (vs. 1) Unilateral procedural justice, Chinese Control variables	0.21**	8.47**	.25	.03	5.04*	9, 114
Equation 5 (vs. 4) Shared procedural justice Unilateral procedural justice, Chinese Control variables	0.28*** 0.16*	16.82***	.30	.05	8.11**	10, 113
Equation 6 (vs. 4) Unilateral procedural justice, foreign Unilateral procedural justice, Chinese Control variables	0.11 0.18*	9.29***	.27	.01	2.16	10, 113
Equation 7 (vs. 6) Shared procedural justice Unilateral procedural justice, foreign Unilateral procedural justice, Chinese Control variables	0.25** 0.08 0.18*	18.64***	.31	.04	6.29**	11, 112

^a $n = 124$. Mean centering was used in equations 3, 5, and 7.

^b Control variables in this table include the six control variables in Table 2 as well as structural uncertainty and cultural distance.

[†] $p < .10$

* $p < .05$

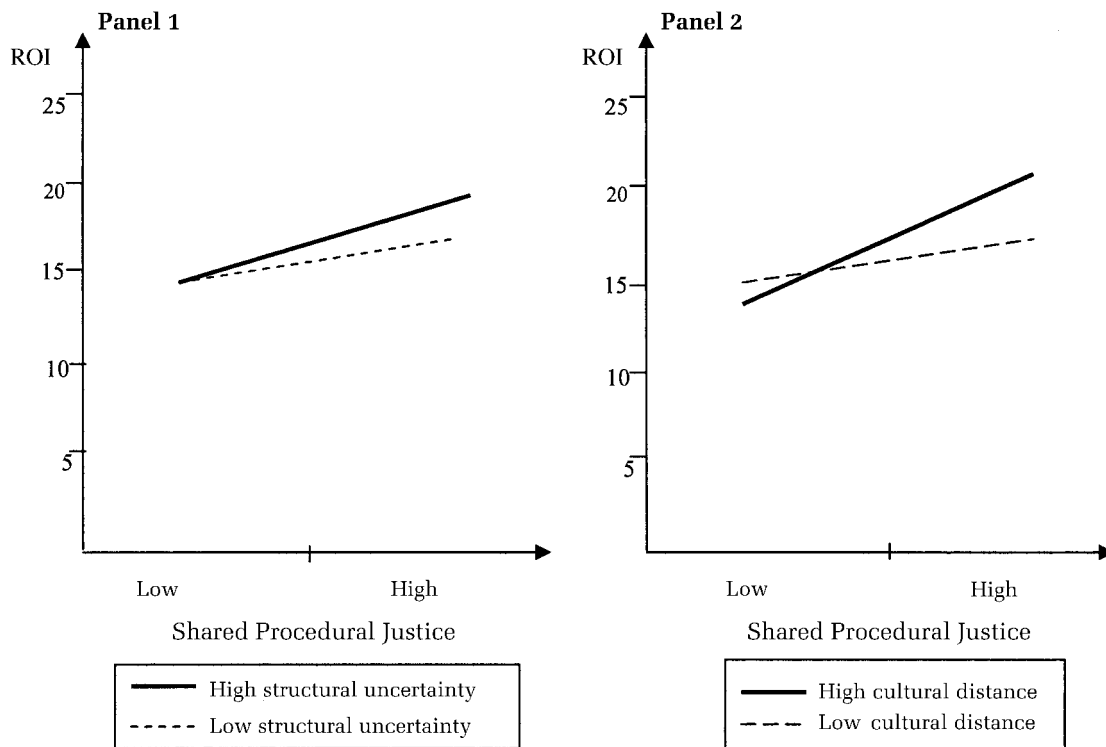
** $p < .01$

*** $p < .001$

($\beta = .28, p < .001$) with Chinese-party-perceived procedural justice ($\beta = .16, p < .05$), also shows that adding shared procedural justice to Chinese-party-perceived procedural justice significantly increases the model's power to predict profitability ($\Delta R^2 = .05$, hierarchical $F = 8.11, p < .01$). In the full model (equation 7), I found that shared procedural justice better predicted variance in profitability than did the combined one-party procedural justice perceptions ($\Delta R^2 = .04$, hierarchical $F = 6.29, p < .01$). These results support Hypothesis 1b—that is, there is a stronger relationship between shared procedural justice and profitability than between unilaterally perceived procedural justice and profitability. It is interesting to note that unilateral justice perceptions are not symmetric for the Chinese and foreign alliance partners when the unilateral perceptions are entered together in a regression: in equation 6, Chinese unilateral justice perceptions are significant, but foreign unilateral

justice perceptions are not; in equation 7, unilateral Chinese perceptions are again significant, but foreign unilateral perceptions are not. Moreover, comparing equation 6 with equation 2 suggests that adding unilateral Chinese justice perceptions significantly increases the predictive power of the model that includes unilateral foreign justice perceptions and control variables ($\Delta R^2 = .03$, hierarchical $F = 4.01, p < .05$). These results suggest that only Chinese justice perceptions stay significant in relation to profitability; foreign justice perceptions are not significant either before or after shared justice perceptions are entered. The possible reason behind this finding is that a local partner's commitment to joint activities is susceptible to its perceived procedural justice, whereas a foreign partner's contribution is either not susceptible or susceptible to a lesser degree. Beamish and Banks (1987) and Madhok (1995) also found asymmetry in that foreign partners tended to tolerate injustice

FIGURE 1
Uncertainty and Culture as Moderators



more than local partners. To further check the above results, I randomly split the sample in half and reran these equations; similar results were obtained.

To test the moderating effects of cultural distance (Hypothesis 2) and structural uncertainty (Hypothesis 3), I incorporated moderated terms into the hierarchical regression analysis; Table 2 shows these results. The hierarchical regressions I performed after controlling for other related variables suggest that structural uncertainty does not independently affect profitability, but its interaction with shared procedural justice has a moderately positive effect on profitability ($\beta = .16, p < .10$, model 4).³ The second moderator variable, cultural distance, demonstrates a similar property: it does not have a direct (main) effect on profitability, but its interaction with shared procedural justice does

and is positive ($\beta = .18, p < .05$, model 4).⁴ Hierarchical *F*-tests confirmed that the predictive power of the full model was significantly stronger after two interaction terms were included (hierarchical $F = 8.19, p < .01$). When each interaction term was added separately, predictive power also significantly increased. These results confirm the importance of the interaction terms in relation to profitability. In addition, I plotted the two interactions in which structural uncertainty (Figure 1, panel 1) and cultural distance (Figure 1, panel 2) are treated as moderator variables affecting shared procedural justice's influence on profitability. Corroborating the moderated regression results, the slopes of the regression lines vary significantly (grow positively steeper) as *Z*-values increase from Z_L (mean minus one standard deviation) to Z_H (mean plus one stan-

³ Whether or not this finding is unique to an emerging economy such as China requires further examination. Structural uncertainty often hinders firm performance (Scherer & Ross, 1990). But industries with high structural uncertainty are typically those in high-growth areas in an emerging economy. This growth effect may cancel out the hazard of uncertainty in the industry's total influence on firm profitability.

⁴ I used Ronen and Shenkar's (1985) country clusters to obtain each foreign party's culture cluster affiliation and further check the cultural effect. The sample had four culture clusters (Anglo, Far Eastern, Germanic, and Nordic) and one independent cluster (Japan). The results in Tables 2 and 3 remained about the same when I controlled for these clusters. When these clusters were added to models 6 and 7 (Table 3), the increase in explained variance (R^2) was not statistically significant.

dard deviation) in both graphs. These results support Hypotheses 2 and 3, suggesting that the positive link between shared procedural justice and alliance profitability becomes stronger when structural uncertainty is high or when cultural distance is great.

DISCUSSION AND CONCLUSION

Procedural justice research is conventionally micro focused and usually scrutinized from a one-sided perspective. In this study, I sought to extend procedural justice research to a macrolevel organizational setting (cross-cultural cooperative alliances) in a new environmental context (an emerging market) and to shared justice judgments perceived by both parties through their boundary spanners, who transform their fairness perceptions into parent firm actions. My analysis suggests that (1) perceptions of procedural justice shared by both parties have a positive relationship with alliance profitability (greater financial returns result when both parties perceive high procedural justice than when both parties perceive low procedural justice), and (2) perceptions of procedural justice shared by both parties have a stronger positive link with profitability than do individual parties' unilateral perceptions of procedural justice (greater financial returns result when both parties perceive high procedural justice than when one party perceives high procedural justice but the other party perceives low procedural justice, even if the average of the two is the same). With controls for certain alliance attributes such as age, size, location, orientation, type, and past performance, my moderated regression analysis demonstrates that shared procedural justice becomes even more important in relation to profitability when cultural distance or structural uncertainty is at a high level. In other words, alliances with greater cultural distance between the investing firms or that operate in industries with greater structural uncertainty experience a stronger positive relationship between shared procedural justice and joint profitability.

The above results have some implications for justice research in addition to alliance research. Justice is an important element in all economic exchanges, but especially in those involving cooperation between two cross-border firms whose repeated exchanges in a long-term alliance agreement are governed by both economic structure and social norms. Because these alliances are loosely coupled systems, the justice judgments of the parties often vary, leading to asymmetry in procedural justice perceptions. It is the looseness in the system that causes differences in procedural justice percep-

tions that often translate into interparty conflicts. It is the coupling aspect of the system that necessitates agreement in procedural justice perceptions and a shared view of fairness in cooperation procedures and exchange processes. The differences in the effects of unilateral views of procedural justice and the effects of shared views of procedural justice should remind future justice researchers that justice consensus has strong implications for aspects of alliance performance, such as profitability. For cooperation aimed at collective achievements, shared procedural justice is an important force for improving joint operations. Theoretically, I argued that shared procedural justice, rather than unilaterally perceived procedural justice, can better improve joint activities via increased relational value, decreased relational risk, reduced relational conflict, and/or improved relational stability. A future empirical effort to test these posited mediators or channels is warranted.

This study provides evidence that shared procedural justice becomes even more important in relation to alliance profitability when an alliance faces great structural uncertainty (a major feature of emerging markets) or when the cultural distance between partner firms is high (a major feature of international cooperative alliances). The above implies that the strength of shared procedural justice in relation to joint profitability is not the same in all circumstances, suggesting that this strength is subject to environmental forces such as industrial uncertainty and cultural heterogeneity. These contingencies support the notion, posited in the loose coupling theory, that sharedness is more essential to coupling when forces for looseness (such as uncertainty and cultural difference) are stronger. It also corroborates justice theory in that procedural justice has a stronger influence on people's attitudes and commitment to joint activities when they perceive greater outcome uncertainty owing to external (e.g., structural uncertainty) or internal (e.g., cultural distance) factors. This is the "compensatory effect" of using a one-way lens. My analysis implies that there may exist another compensatory effect: interparty agreement in procedural justice perceptions may strengthen the relationship between procedural justice and joint gains, going beyond the effect of procedural justice itself, and this augmentation seems to be even more evident when joint activities involve greater outcome uncertainty associated with industrial structure and cultural differences.

Despite procedural justice's critical role in alliance growth, very few studies have used it to explain the evolution of cooperative alliances, an organizational form used today by a growing number

of firms for their corporate diversification and global expansion. In an alliance setting, boundary spanners from both parties perceive procedural justice; thus it is dyadic in nature and requires diagnoses of shared judgments from both parties. In my view, shared procedural justice perceptions can affect the process as well as the structure of interpartner exchange, two central concepts in alliance research. If both parties view the process itself as fair, it will add more relational value from increased mutual commitment and sustained cooperation. It will also complement the exchange structure by alleviating relational risk or relational uncertainty arising from unilateral opportunism and pursuit of private interests. It is interesting to note that cultural distance and structural uncertainty, two main variables extensively examined by researchers on cross-cultural alliances as hindrances to alliance stability, were found to moderate the link between shared procedural justice and alliance profitability. Alliance age was also found to have a significant and positive impact on profitability, validating the importance of time-related experience in alliance growth, as held by alliance theory. Thus, it is firm-specific familiarity with a local environment, not the origin of a foreign investor (nation-level cultural distance was not found to have a main effect on profitability), that matters to alliance profitability.

What this study did not and was not designed to address is this: Under what conditions will different parties' justice perceptions converge, and how can they reduce disagreement in these perceptions? These are important research questions because the ultimate goal is to heighten shared justice judgments between boundary spanners. Moreover, this study emphasized procedural justice only; an extended investigation might study whether shared perceptions of other types of justice (for instance, distributive fairness and interactive fairness) also have a larger effect on cooperation outcomes than unilateral perceptions in these areas. In addition, my finding that cultural distance did not have a main effect on profitability should be interpreted with some caution. Given the limitations of cross-sectional data, I can only conclude with certainty that there is no significant difference in profitability among sample alliances established by firms with different origins. Scores on cultural dimension indexes in Hofstede's (2001) data set may not represent the actual cultural attributes of foreign firms in my sample or reveal these firms' actual efforts to culturally adapt to host country markets.

Methodologically, I caution that the above results were derived from an analysis of sample alliances concentrated in a single province in a single coun-

try and that alliance performance was measured by profitability only. Future research should use a more representative sample of firms located in different countries and should define alliance performance by a broader range of outcome indicators containing both objective measures (e.g., sales growth) and subjective ones (e.g., parent satisfaction). Future research should also use longitudinal data to better assess how shared procedural justice affects social and organizational interactions between partner firms over time. Longitudinal data may also allow examination of a reciprocal causal relationship between procedural justice and corporate performance. Although I used lagged financial information and controlled for past performance, my data set was insufficient to depict the reciprocal causal link whose existence has been richly documented in the field of organizational justice. This causality can be further unveiled by investigating the channeling effect of posited mediators I conceptually discussed, such as reduced relational risk, heightened relational value, and increased interparty conformity on alliance outcomes. Using path analysis, future research can verify whether shared procedural justice contributes more to joint returns than unilaterally perceived procedural justice through these channels. Finally, it would be worthwhile to take further steps to discern to what extent shared perceptions of varying aspects of procedural justice, such as bias suppression, consistency, and representativeness, have disparate influences on firm performance. Also valuable would be further research on the extent to which these influences are altered by social and organizational variables such as attachment, trust, teamwork, and familiarity.

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